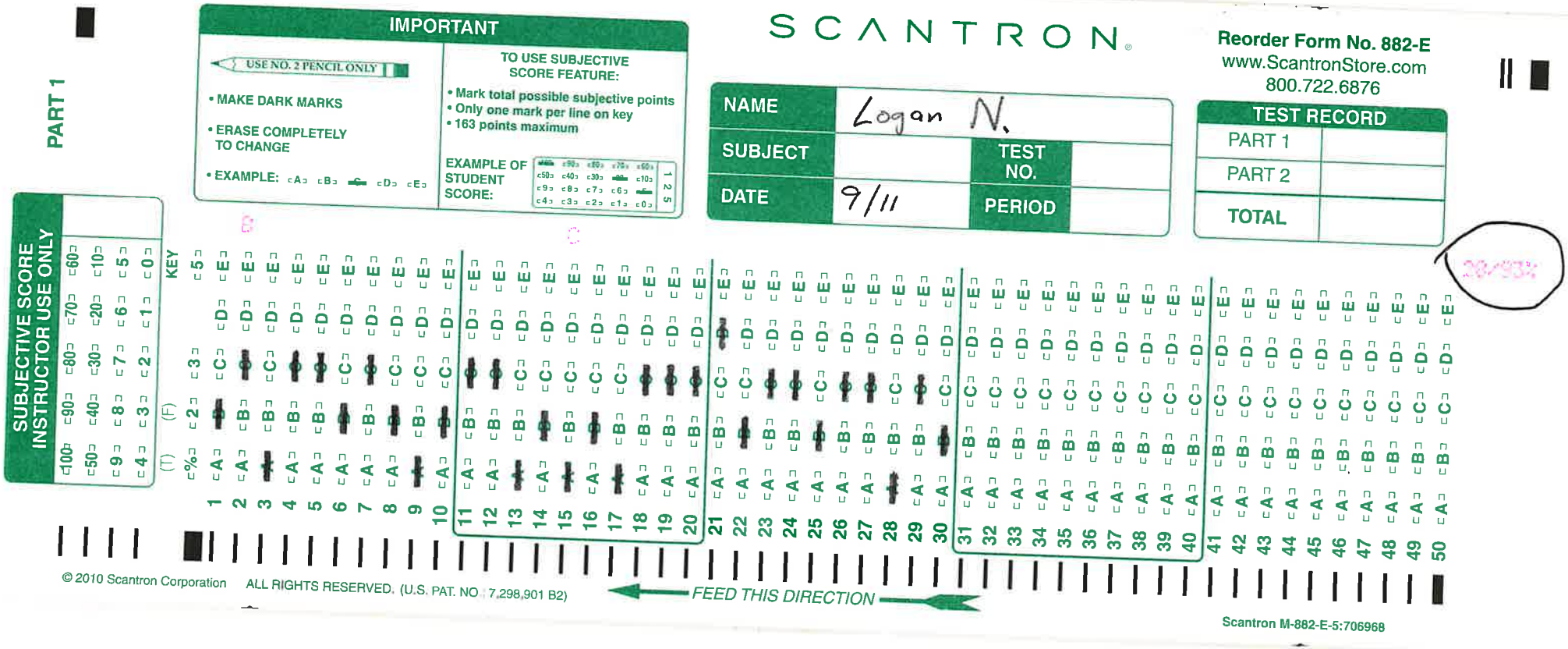

IMPORTANT
USE NO. 2 PENCIL ONLY
• MAKE DARK MARKS
• ERASE COMPLETELY TO CHANGE
• EXAMPLE:
TO USE SUBJECTIVE SCORE FEATURE:
• Mark total possible subjective points
• Only one mark per line on key
• 163 points maximum
EXAMPLE OF STUDENT SCORE:
SCANTRON
Reorder Form No. 882-E
www.ScantronStore.com
800.722.6876
NAME
Logan N.
SUBJECT
TEST NO.
DATE
9/11
PERIOD
TEST RECORD
PART 1
PART 2
TOTAL
PART 1
SUBJECTIVE SCORE INSTRUCTOR USE ONLY
KEY
(T)
(F)
© 2010 Scantron Corporation ALL RIGHTS RESERVED. (U.S. PAT. NO. 7,298,901 B2)
FEED THIS DIRECTION
Scantron M-882-E-5:706968

PART 2

NAME

	(T)	(F)		KEY
	%	2	3	5
51	A	B	C	D E
52	A	B	C	D E
53	A	B	C	D E
54	A	B	C	D E
55	A	B	C	D E
56	A	B	C	D E
57	A	B	C	D E
58	A	B	C	D E
59	A	B	C	D E
60	A	B	C	D E
61	A	B	C	D E
62	A	B	C	D E
63	A	B	C	D E
64	A	B	C	D E
65	A	B	C	D E
66	A	B	C	D E
67	A	B	C	D E
68	A	B	C	D E
69	A	B	C	D E
70	A	B	C	D E
71	A	B	C	D E
72	A	B	C	D E
73	A	B	C	D E
74	A	B	C	D E
75	A	B	C	D E
76	A	B	C	D E
77	A	B	C	D E
78	A	B	C	D E
79	A	B	C	D E
80	A	B	C	D E
81	A	B	C	D E
82	A	B	C	D E
83	A	B	C	D E
84	A	B	C	D E
85	A	B	C	D E
86	A	B	C	D E
87	A	B	C	D E
88	A	B	C	D E
89	A	B	C	D E
90	A	B	C	D E
91	A	B	C	D E
92	A	B	C	D E
93	A	B	C	D E
94	A	B	C	D E
95	A	B	C	D E
96	A	B	C	D E
97	A	B	C	D E
98	A	B	C	D E
99	A	B	C	D E
100	A	B	C	D E

FEED THIS DIRECTION

8 10054 00005 1

SAN JUAN COLLEGE

gm asep

Unit 2-Manual Transmissions/Transaxles

Logan

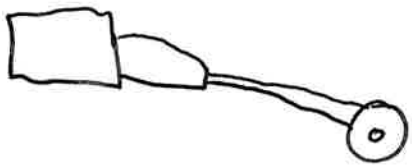
1. Shift interlocks are used:
   a. To insure that reverse gear operates smoothly.
   b. To prevent two gears from engaging at the same time.
   c. To prevent high speed downshifts
   d. Help hold the shift rails in gear

got 2 wrong

2. During a manual transmission normal up shift the synchronizer blocker ring must. (A) Speed up the main shaft (output shaft) assembly. (B) Slow down the clutch disc, main drive gear, cluster gear, and speed gears. Which is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B

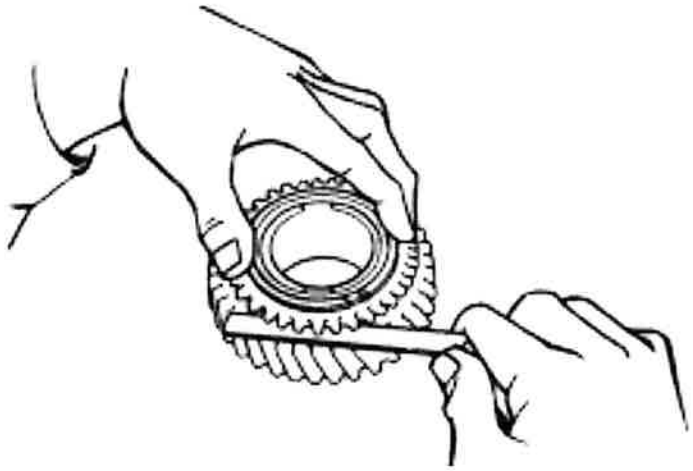
b was correct

3. Two technicians are discussing a transmission that is noisy in first, second, third and fifth but quiet in fourth gear. Technician A says that a bad input shaft gear could cause this. Technician B says this could be caused by a faulty 4th gear speed gear. Who is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B



4. Two technicians are discussing the operation of Synchronizer\Blocker rings. Technician A says that they equalize the speeds of the synchronizer assembly and the speed gear being engaged. Technician B says they block the synchronizer sleeve briefly during a shift. Who is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B

5. Technician A says that an improperly adjusted shift linkage can cause a transmission to jump out of gear. Technician B says that a faulty clutch can cause hard shifting. Who is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B



6. The check being performed is used to:
   a. Determine shift key wear
   b. Blocker ring wear
   c. Speed gear thrust clearance
   d. Dog teeth wear

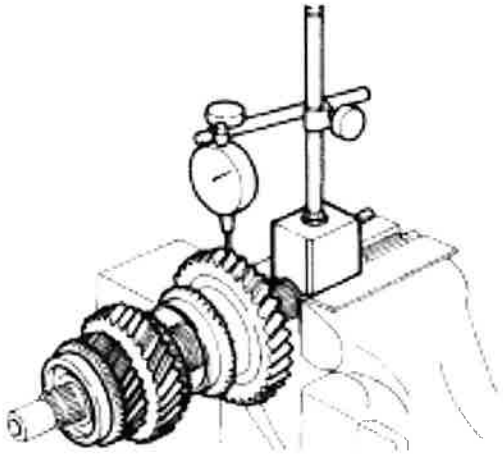
7. Technician A says that each speed gear should have a specified amount of end play (axial) clearance. Technician B says that axial clearance can be adjusted on some manual transmissions\transaxles with select fit thrust washers or snap rings. Who is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B

8. Oil grooves on a synchronizer blocker ring are used to (A) give a visual gage as to the amount of blocker ring wear. (B) cut through the film of oil that may exist on the speed gear. Which is correct?
   a. A only
   b. B only
   c. Both A and B
   d. Neither A nor B

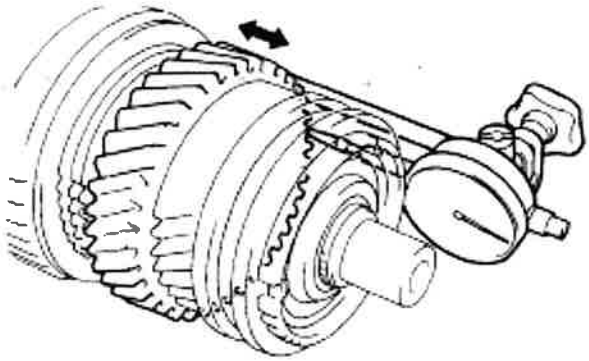
I chose b b is correct

This is a question I had to think about but its not that complex

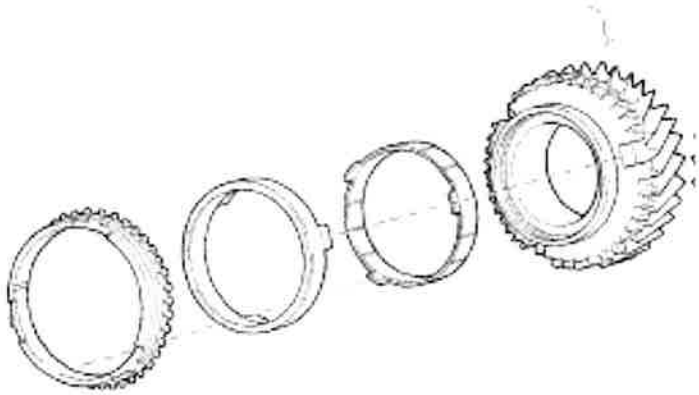
9. The inspection being performed is used to:
   a. Measure radial clearance
   b. Measure axial clearance
   c. Measure input shaft run out
   d. Measure the synchronizer wear



10. The inspection being performed is used to:
   a. Measure radial clearance
   b. Measure axial clearance
   c. Measure input shaft run out
   d. Measure the synchronizer wear



11. The illustration shown is a:
   a. Shift Fork assembly
   b. Single Cone Synchronizer
   c. Triple Cone Synchronizer
   d. Shift Shaft assembly



This stuff is easy

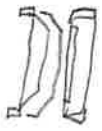
12. Manual Transmission lubrication is being discussed: Technician A says that some Manual transmissions should not be flat towed due to lack of lubrication. Technician B says that most Manual transmissions use splash lubrication. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

13. Two technicians are discussing oil viscosity Technician A says that oil with the wrong viscosity could cause a shifting concern. Technician B says that the wrong oil will not affect shift effort. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

14. A transmission grinds while shifting into third gear only. Technician A says that the most likely cause is a faulty clutch assembly. Technician B says the cause is most likely the third gear synchronizer assembly. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

15 wrong
C correct

15. A manual transmission pops out of 5th gear. Technician A says that excessive 5th gear axial clearance could be the cause. Technician B says that an excessively worn synchronizer assembly may be the cause. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B



couldn't decide how a worn synchronizer could pop a gear out, but I do think I got it now.

16. A manual transmission equipped vehicle does not move in any gear except 4th. Technician A says that the transfer case is probably shifted into neutral. Technician B says that a damaged input gear set is the most likely the caused for this concern. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

17. Technician A says that one of the major differences between a manual transmission and a manual transaxle is the addition of the final drive unit in the manual transaxle. Technician B says that synchronizer assemblies in a manual transaxle function different when compared to a manual transmission. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

18. The transaxle final drive, drive-pinion is located _____.
    a. On the input shaft
    b. On the clutch shaft
    c. On the output shaft
    d. On the shift shaft

19. A transaxle has overheated and sustained internal damage. What is the most likely cause?
    a. The driver was slipping the clutch
    b. Worn synchronizer sleeve(s)
    c. A lack of or low lubrication oil level
    d. Defective input shaft bearings

20. In a transaxle, the synchronizer assemblies are mounted on the (A) input shaft, or (B) output shaft. Which is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

21. The inboard CV joints on the drive-shafts (half-shafts) are splined to the _____.
    a. Differential pinion gears
    b. Output shaft
    c. Differential case
    d. Differential side gears

22. The final drive ring and pinion gears in a typical transaxle are _____.
    a. Hypoid gears
    b. Helical gears
    c. Spiral bevel gears
    d. Straight cut gears

23. Technician A says that a worn transaxle shift fork can cause the transaxle to jump out of gear. Technician B says that if the driver rests his/her hand on the shift lever this will increase the amount of wear on the shift fork. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

24. All of the following conditions can cause hard shifting ***EXCEPT***:
    a. Dragging clutch
    b. Wrong trans lubricant
    c. A chipped speed gear tooth
    d. Worn Synchronizer rings

25. Two technicians are discussing a transaxle that is noisy in second but quiet in all other gear ranges. Technician A says that a damaged final drive pinion gear could cause this. Technician B says this could be caused by a damaged second speed gear. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

26. The owner of a vehicle equipped with a manual transaxle complained that gear clash was heard whenever shifting. Which is the ***least*** likely to be the cause?
    a. Worn blocker ring
    b. Low hydraulic clutch fluid level
    c. Defective output shaft bearing
    d. Warped clutch disc

27. Technician A says some transaxles use steel blocker rings with a Kevlar lining. Technician B says some transaxles use triple cone blocker rings. Which tech is correct?
    a. Tech A
    b. Tech B
    c. Both tech A and B
    d. Neither tech A nor B

28. Which of the following could be the cause of a transaxle noise heard when cornering but not when driving straight?
    a. Damaged differential gear
    b. Worn output shaft pinion drive gear
    c. Damaged final drive ring gear
    d. Broken 5th gear blocker ring

29. A vehicle equipped with a manual transaxle has a customer's complaint of a gear whine when driving in second gear. Technician A says that improper second gear tooth contact may be the cause. Technician B says that excessive radial gear clearance can be the cause of improper gear tooth contact. Who is correct?
    a. A only
    b. B only
    c. Both A and B
    d. Neither A nor B

30. Technician A says that when making a turn to the left, the left wheels will turn faster than the right. Technician B says a basic differential will apply the most torque to the wheel with the least amount of traction. Which tech is correct?
    a. Tech A
    b. Tech B
    c. Both tech A and B
    d. Neither tech A nor B



End